AURORA Proteus Run and Test Guide

How to run the simulation and verify expected behaviour

Before Running

- Confirm the BMS, VCU and Motor Controller devices are each loading the newest compiled ELF file.
- Set all three human-readable Virtual Terminals to 115200 baud, 8 data bits, no parity, 1 stop bit.
- Keep all ECU-to-ECU UART links at 9600 baud.
- Confirm VCU PA9 is connected to Motor PA3 and Motor PA2 is connected to VCU PA10.
- Confirm BMS PA2 is connected to VCU PA3.
- Confirm all sections share the same GND reference.

Expected Startup

Screen	Expected boot text
BMS	AURORA BMS READY
VCU	AURORA VCU READY
Motor Controller	AURORA MOTOR CTRL READY

Test Scenarios

Action	Expected BMS	Expected VCU	Expected Motor Controller
Normal drive: ignition ON, ACC ≈ 50%, no brake/fault	NORMAL; limit 100%	DRIVE; command ≈ 50%; DTC 0	DRIVE; RX D/50; PWM ≈ 50%; DTC 0
Increase BMS temperature into WARNING	WARNING; limit 80%	DRIVE permitted with reduced limit	PWM follows reduced VCU command
Increase BMS temperature into DERATE	DERATE; limit 50%	Final command is reduced by the 50% limit	PWM follows reduced command
Increase BMS temperature to SHUTDOWN	SHUTDOWN; limit 0%	FAULT; motor command 0	STOP / PWM 0
Apply brake	No change unless temperature changes	REGEN; propulsion command removed	REGEN indication; propulsion PWM 0
Activate motor-fault switch	No change	MOTOR FAULT / DTC 1; propulsion removed	FAULT active
Connect charge switch	No change	CHARGE; drive disabled	STOP / PWM 0

Proteus Performance Note

The simulation can run much slower than real time because Proteus is simultaneously executing three STM32 models, UART traffic, ADC behaviour, PWM switching, the motor power stage and several Virtual Terminals. This affects how fast the simulation clock advances, not the intended control relationships.

- Hardware PWM is used instead of software PWM to reduce simulation load.
- Debug screens update slowly by design to reduce terminal rendering overhead.
- Remove temporary debug terminals after troubleshooting.
- If SPICE reports 'timestep too small', inspect the motor/MOSFET power stage, flyback path, grounding and non-ideal series resistance.

Known Limitations

- No dedicated battery-current sensor is implemented in the current Proteus schematic.
- No physical regenerative energy return is implemented in the single-MOSFET motor stage.
- The BMS in Proteus is temperature-focused; SOC is handled as a VCU input for the demonstration.
- UART is used for the prototype; a production-style EV implementation would typically use an automotive network such as CAN.